Calculus and vector

Unit-I ① If $y = e^{\tan^{-1}x}$, prove that

$$(1+x^2)y_{n+2} + [2(n+1)x - 1]y_{n+1} + n(n+1)y_n = 0$$

② If $y = (1-x)^{-b} e^{-bx}$, prove that

(i) $(1-x)y_1 = bxy$

(ii) $(1-x)y_{n+1} - (n+bx)y_n - nby_{n-1} = 0$.

③ If $y = \cos(m\sin^{-1}x)$, prove that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0$$

④ If $y = a\cos(\log x) + b\sin(\log x)$, show that

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0.$$

⑤ If $y = (\sin^{-1}x)^2$, prove that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0$$

⑥ If $y = \left[x + \sqrt{1+x^2}\right]^m$, prove that

$$(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0$$

⑦ If $y = e^{m\cos^{-1}x}$, prove that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2+n^2)y_n = 0$$

⑧ If $y^{1/m} + y^{-1/m} = 2x$, prove that

$$(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0$$

(2)(1) using Maclaurin's theorem, Expand (Show that)

(i) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

(ii) $e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$

(iii) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$

(iv) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

(v) $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots$

(vi) $\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots$

(vii) $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots$

(viii) $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots$

(2) using Taylor's theorem, find the value of $\tan 46^\circ$ correct to four decimal places.

(3) prove that
$\log(1+e^x) = \log 2 + \frac{1}{2}x + \frac{1}{8}x^2 - \frac{1}{192}x^4 - \cdots$

(4) Expand $\cos x$ in power of $\left(x - \frac{\pi}{4}\right)$ upto 4 terms

(5) using Taylor's series find the value of $\cos 64^\circ$ correct to four decimal places.

(6) obtain the maclaurin's expansion of $\tan\left(\frac{\pi}{4} + x\right)$ and hence obtain the value of $\tan 46'30'$ to four decimal places.